`olmoearth_inferenceX`: confidence-based review, comparison and accuracy estimation for Earth-observation classification maps

Technical report, version 1.9.0

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https://github.com/2imi9/olmoearth_inferenceX
https://olmoearth-inferencex.readthedocs.io

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Abstract

This report describes `olmoearth_inferenceX`, a Python package for assessing classification maps from Earth-observation models, and the record behind it. Without labels, the package ranks a map's windows for review by the model's confidence, explains each flagged window by label-free cues, and compares two maps with regard to their dates. With a labelled sample, it estimates the error rate and per-class accuracy with design-based intervals, certifies a high-confidence zone whose error rate is bounded with a stated probability, and, from labels drawn where two maps differ, gives an interval on their accuracy difference that names the more accurate map when it excludes zero. Read through linear probes on OlmoEarth Base's frozen embeddings, the confidence margin closes a median 0.68 of the gap between a random and a perfect review order on the 24 tasks of the model authors' embedding suite, and a review of the least confident 10% holds a median 0.214 of the errors, about twice a random 10%; it beats a random order in all 3,560 cells of sixteen encoders, their tasks and ten probe seeds, and beats informative no-model controls on three references produced outside this project. On PASTIS, a probe read without the optical input it was trained with erred confidently (59.8% of its errors as confident as the median correct window, against 6.0%), though not on CropHarvest China 6; given a raster of input conditions, the package therefore ranks, samples and certifies each condition on its own. The random-sample interval is exact; it remains valid when a reviewer cannot judge some windows, and widens to sharp bounds, proved in Lean, for a reviewer whose error rates the user states. The certified zone met its guarantee on all 110 graded cells. Over 2,514 pairs of encoder maps, 100 labels drawn among the windows where two maps differ named the more accurate map on 60.3% of draws, against 12.7% for 100 labels drawn from the whole map; on the 1,709 pairs that differ on more than 100 windows, where the labels are a sample rather than all of them, 42.8% against 11.3%. On two published land-cover products graded against an independent probability sample of 25,000 plots, the error-rate interval held its coverage, labels drawn where the products differ never named the one less in agreement with the reference (they named one on 6% to 29% of draws at 50 to 200 labels), and LCMAP's own confidence ranked its errors with an AUROC of 0.75; certified zones on that confidence met their guarantee in every cell, and at 300 labels the default rule returned no zone on 87% of draws at α half the error rate, though 65% of the plots lie in a zone within it. The record comprises 92 experiment scripts and 228 ledger claims, 226 of them checked against a committed artifact, and reports failed predictions as failed.

1 Introduction

A classification map from an Earth-observation model is usually applied where no reference labels exist. This report addresses four questions: which windows to review first when only a fraction can be checked; how two maps of one area differ, when they come from different crops, backbones,

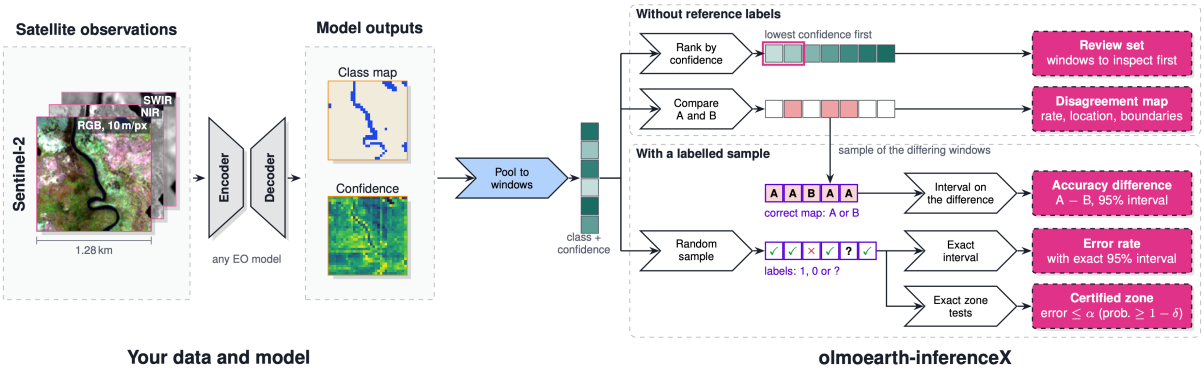


Figure 1: The structure of the package. It reads a model's class scores (and optionally a second map and a raster of input conditions), pools them to windows that each carry a class and the model's confidence, and answers two kinds of question. Without labels, it ranks the windows by confidence into a review set (Section 2.1) and compares two maps (Section 2.2). With a labelled sample, windows drawn at random give the error rate with an exact interval (Section 2.3) and, by exact tests on the most confident zones, a certified zone (Section 2.4); windows drawn where two maps differ give the accuracy difference (Section 2.5). Each step runs per input condition when a condition raster is given.

sensors, encoders, training states or dates; once some windows are labelled, how wrong the map is, how accurate each class is, and which part of it can be used with a stated error bound; and which of two maps is more accurate, from as few labels as possible.

`olmoearth_inferenceX` ranks 4×4-pixel windows by the model's confidence margin with label-free cues (Section 2.1), compares two maps with regard to their dates (Section 2.2), estimates error rates and per-class accuracies from a labelled sample (Section 2.3), certifies a zone of the most confident windows (Section 2.4), and estimates the accuracy difference of two maps from labels where they differ (Section 2.5). It reads arrays and rasters and loads no model, and it serves the same commands to a language-model agent (Section 2.6); Figure 1 shows the four answers. In the evaluation the label-free readings are graded against labels and never fitted to them [labels-grade-never-train].

Relation to prior work. No component is new in isolation. Risk-coverage evaluation comes from selective classification [Chow, 1970, Geifman and El-Yaniv, 2017, Geifman et al., 2019], softmax confidence as the error-detection baseline from Hendrycks and Gimpel [2017] and Jaeger et al. [2023], and the margin from margin sampling [Scheffer et al., 2001, Lewis and Gale, 1994]; related measures are Rottmann et al. [2020] and Meyer and Pebesma [2021]. The estimators follow design-based map-accuracy practice [Olofsson et al., 2014, Stehman and Foody, 2019], with the cluster-sampling correction of Stehman [1997]; stratifying a classifier's evaluation sample by its confidence was proposed by Kumar and Raj [2016], and prediction-powered inference [Angelopoulos et al., 2023a], including a design-based spatial form [Shirota, 2026], addresses the same estimation task. The certified zone follows risk control by exact tests [Bates et al., 2021, Angelopoulos et al., 2025]. Disagreement maps between land-cover products [Fritz and See, 2008] and paired accuracy tests on a shared sample [Foody, 2004] precede the comparison of two maps, and the bounds for a reviewer who errs are the partially identified form of the correction of Rogan and Gladen [1978]. Lehmann et al. [2026] report that geospatial foundation models grow overconfident under corruption and that confidence-based abstention cannot defer around their confidently wrong predictions; apart from exp88's missing input (Section 4.6), this record has not applied such shifts. This work joins these parts in one after-the-fact tool for maps from Earth-observation foundation models and tests the known signals against the model's confidence

and against controls that do not use the model.

2 Methods

2.1 Windows, the confidence signal and the ranking

Pixels are grouped into windows of 4×4 pixels, the unit of ranking, comparison and estimation. A window's decision is the majority class of its pixels, and a tie goes to the class whose voting pixels are more confident. For a pixel with logits $z_{(1)} \geq z_{(2)} \geq \ldots$, the confidence is the margin

$$m = z_{(1)} - z_{(2)}, \tag{1}$$

and for a binary logit map the absolute logit $|z|$. For a probability map it is the top class probability, which ties where probabilities saturate. The option `form="top1"` ranks a multi-class logit map by the window mean of $\log p_{(1)}$, computed from the logits without ties. Otherwise a window's confidence $m(w)$ is the pixel confidence averaged over the window, and its suspicion is $\sigma(w) = -m(w)$. Except exp89, which graded the package's top-1 form on single points, the experiments of Section 4 graded the margin between the two highest window-mean class probabilities; the package's per-pixel forms on windows were graded on the seven segmentation tasks only [window-aggregator-is-not-the-lever].

A window is a boundary window when any of its eight neighbours has a different decision. The confidence order sorts windows by decreasing suspicion; the boundary-first order takes the boundary windows first, each group by suspicion. The review set at budget b is the first $\max(1, \mathrm{round}(bN))$ of the N valid windows. Each flagged window carries the cues it satisfies: boundary, low confidence (the least confident fifth), tiling instability (a changed decision under a two-pixel crop shift) and, for water maps, spectral ambiguity. A cue's enrichment, its share among error windows over its share among correct windows, is quoted as measured on labelled data; since version 1.3.0 the boundary cue is quoted with the range measured on the suite and the map's own boundary prevalence.

Input conditions. A map can be read from different inputs in different places, for instance radar alone under cloud and radar with optical imagery elsewhere. Given an integer raster of input conditions on the map's grid (`--condition`, since version 1.4.0), `assess` writes a review set per condition, and `sample`, `estimate` and `certify` treat each condition as its own population. Without it, the output notes that confidence ranked across input conditions can place a model's confident errors late (Section 4.6).

2.2 Comparison of two maps

For hard window decisions A and B of two maps on one grid, over the windows V both predict, the package reports without labels the disagreement rate $|\{w \in V : A(w) \neq B(w)\}|/|V|$, pooled and per group (tile, event or zone), the enrichment of each cue among differing windows, and the ϕ coefficient between two disagreement masks, which measures whether two differences occupy the same windows. A rate is read against the reseed floor, the rate at which refitting the head with another seed changes decisions. With a full label raster, a separate block adds the cross-tabulation of the two error maps and the share of differing windows on which each map matches the label.

Each map may carry the date or period it represents. At one time a difference implies an error in at least one map; across times it may be change on the ground, and the package then grades which map is right only when the labels' date is given, reporting which map the labels match in time; with one date stated, grading is refused. Without labels the package does not decide which map is right; with a sample of labels it says which is more accurate (Section 2.5).

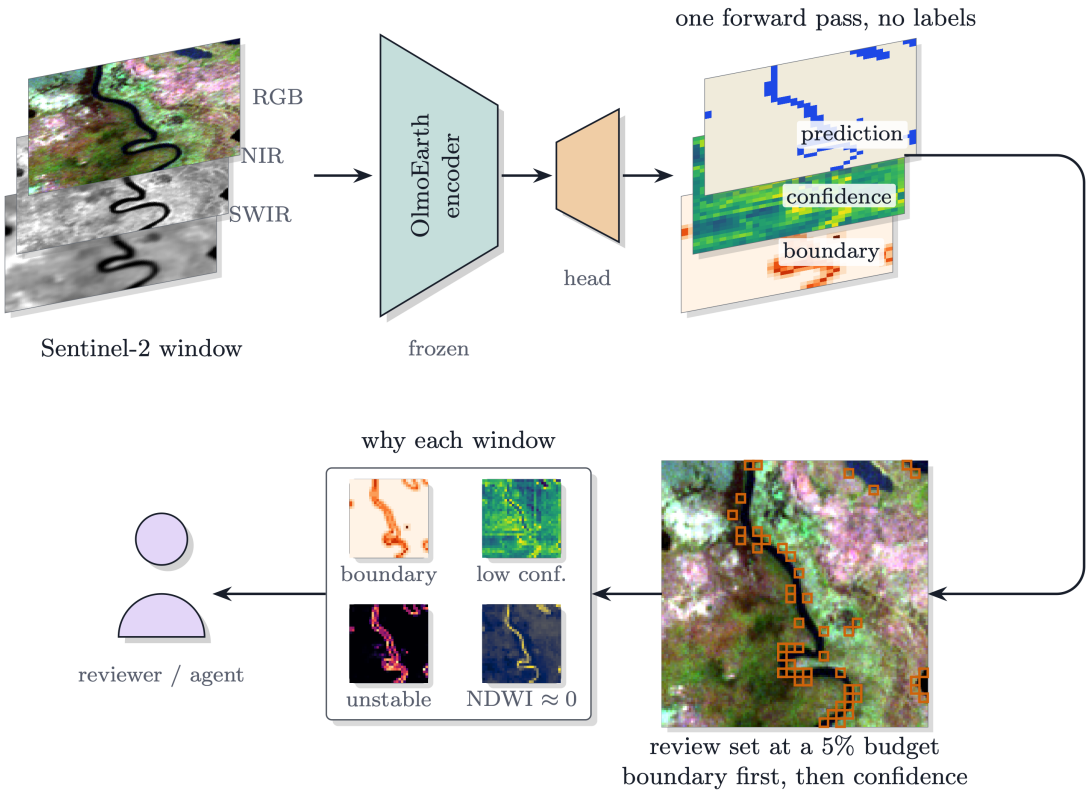


Figure 2: One scene through the label-free assessment: Sentinel-2 bands, the frozen OlmoEarth encoder and a task head, the prediction with its confidence and boundary layers, the review set at a 5% budget, and the cues per flagged window.

2.3 Design-based estimation

Let the map have N valid windows, $e_i = 1$ where the map's class differs from the reference class, and $\theta = N^{-1} \sum_i e_i$. A reviewer labels B windows under one of five designs.

Confidence-stratified design (default). Five strata are cut at the quintiles of the margin, with sizes N_h and weights $W_h = N_h/N$. The budget follows Neyman's allocation, which minimises the variance of the stratified estimate at a fixed budget,

$$n_h = B \frac{N_h S_h}{\sum_j N_j S_j}, \qquad S_h \approx \sqrt{q_h(1 - q_h)}, \tag{2}$$

where q_h is the mean of $1 - p_{(1)}$ in stratum h, floored at 0.02, and each stratum receives at least two labels. With p_h the error share among stratum h's labels and $f_h = n_h/N_h$,

$$\hat{\theta} = \sum_h W_h \, p_h, \qquad \hat{v} = \sum_h W_h^2 (1 - f_h) \, \frac{p_h(1 - p_h)}{n_h - 1}, \tag{3}$$

both unbiased under stratified sampling without replacement, with Wilson's interval at the effective sample size $\hat{\theta}(1-\hat{\theta})/\hat{v}$, after the effective-sample-size approach of Korn and Graubard [1998]. This interval is nominal: it has no finite-sample guarantee, and its coverage is known from simulation only. The proportional design uses the same strata, allocates the budget by stratum size, and has the same estimator and interval.

Simple random design. The sampled error count k is hypergeometric, and the interval for $\theta = K/N$ holds every K whose two tail probabilities at k both exceed 0.025, widened if needed to contain k/n; its coverage is at least 95% for every (N, K, n). The package has printed this exact

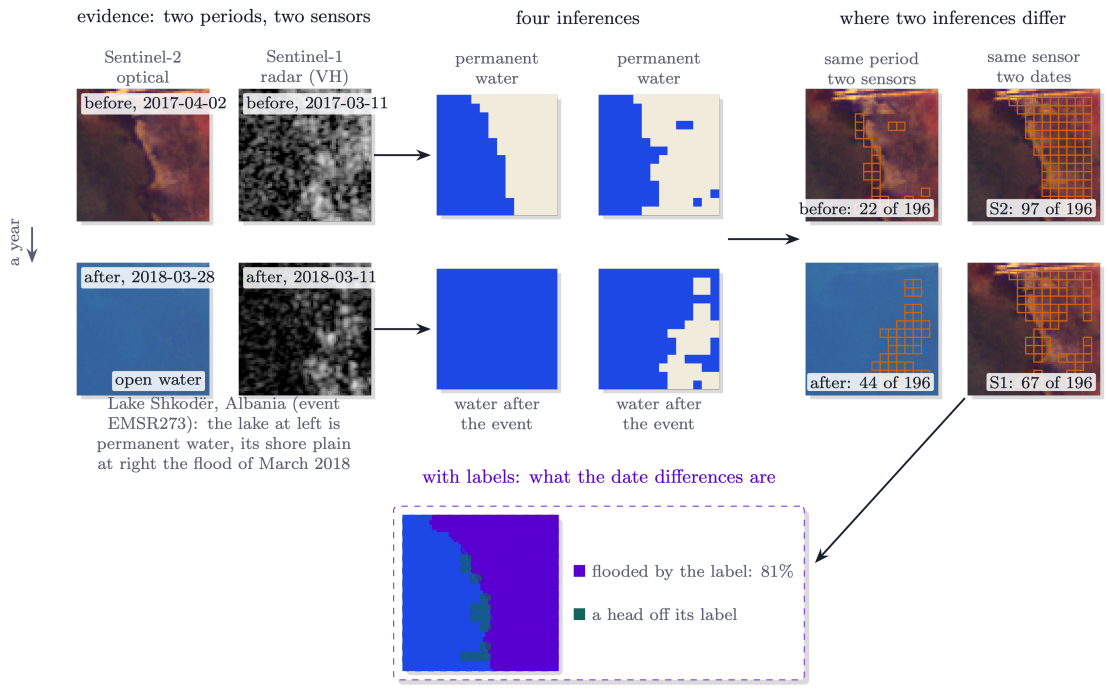


Figure 3: Four dated inputs of one GEOID-Flood chip, each read on identical windows: the same-period and same-sensor differences on the scene, with the comparison against labels shown separately.

interval since version 1.3.0, in place of the finite-population Wilson interval that the experiments of Section 4.11 graded.

Condition design. With L input conditions, the budget is split equally across them, a condition too small for its share being labelled in full, and the model's confidence is not read. Each condition c gets the exact interval $[\ell_c, h_c]$ at 95%. The whole-map interval is $[\sum_c W_c \ell'_c, \sum_c W_c h'_c]$, where $[\ell'_c, h'_c]$ is condition c's exact interval at $1 - 0.05/L'$, L' counts the conditions not labelled in full, and the sums are taken in exact fractions. By the union bound it covers at least 95% for every population and allocation. A stratified Wilson interval with the conditions as strata is not used, because it can cover far less when a large clean condition sits beside a small degraded one.

Tile design. Tiles are taken in a random order and up to 16 windows labelled in each until the budget is met, from at least five tiles. With n_i the valid windows of sampled tile i, $\bar{y}_i$ the error share among its labels and $\bar{n}$ the mean of the n_i, the estimate is the ratio $\hat{\theta} = \sum_i n_i \bar{y}_i / \sum_i n_i$ with ultimate-cluster variance $\sum_i n_i^2 (\bar{y}_i - \hat{\theta})^2 / (t(t-1)\bar{n}^2)$ over t tiles and a t quantile, reported beside the independent-sample interval; its variance ratio to simple random sampling is the design effect. This interval has no guarantee.

Windows the reviewer cannot judge. A reviewer may enter `?` for a window that cannot be judged and keep its row. With u such windows among the labels, the interval runs from the lower end computed with every `?` counted right to the upper end computed with every `?` counted wrong; under a random sample this is $[L(k), U(k+u)]$. Because both ends of the exact interval move one way with the error count, the result contains the interval the true labels would give, so it covers at least 95% however the unjudged windows were chosen, including by an adversary who sees the truth. Under the confidence and proportional designs any `?` switches the interval to the union bound over strata, and most of the extra width then comes from that switch. The tile design and per-class accuracy refuse `?`.

A reviewer who errs. Let e_0 be the share of truly correct windows that the reviewer marks wrong and e_1 the share of truly wrong windows that the reviewer passes. The error rate p of the

reviewer's labels then satisfies $p = \theta(1 - e_1) + (1 - \theta)e_0$. If the user states bounds $e_0 \le E_0 < 1$ and $e_1 \le E_1 < 1$, then

$$\max\Big(0, \frac{p - E_0}{1 - E_0}\Big) \;\le\; \theta \;\le\; \min\Big(1, \frac{p}{1 - E_1}\Big), \tag{4}$$

and both ends are attained, so the bounds are sharp. `estimate --reviewer-false-alarm` E_0 `--reviewer-miss` E_1 maps the interval's ends $[L, U]$ through these increasing functions, so the widened interval covers θ whenever $[L, U]$ covers p; the rates are the user's statement and are not measured. The bounds, their sharpness, the transfer of coverage and the step from monotone interval ends to the `?` interval are proved in Lean 4 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]; the proofs were archived from the main branch on 10 October 2026 and are in the `lean/` folder at tag `v1.9.0` (`https://github.com/2imi9/olmoearth_inferenceX/tree/v1.9.0/lean`). The proofs do not cover the monotonicity of the package's own intervals, which the tests check numerically, and the continuous integration does not build them.

Review-set check. Labelling the review set gives the error share capture(b) θ/b, where capture(b) is the set's share of all errors. Windows labelled without a design are accepted as a random sample only if their mean suspicion percentile is no more than four standard deviations above 0.5, a one-sided test for a review set's enrichment. Passing the check does not make labels a random sample: labels from the confident end of a map, or clustered by tile, pass it.

Per-class accuracy. With map class m and reference class r, the user's accuracy is $\mathrm{UA}_c = \#\{m{=}c, r{=}c\}/\#\{m{=}c\}$, the producer's accuracy $\mathrm{PA}_c = \#\{m{=}c, r{=}c\}/\#\{r{=}c\}$ and the error-adjusted share $A_c = \#\{r{=}c\}/N$. Under a random sample UA_c has the exact interval, and A_c and PA_c [Olofsson et al., 2014] have Wilson intervals at the effective sample size; under the confidence design each is a ratio of Horvitz–Thompson (inverse-probability-weighted) totals with a linearised variance. Warnings mark classes with fewer than 30 labels, thin or near-census sampling, one to four sampled errors, or no prediction.

2.4 The certified zone

Windows are ordered by decreasing margin, ties by index; the zone at coverage c is the first $n_c = \mathrm{round}(cN)$ windows, and its risk $R(c)$ is the error rate inside it, for $c \in \{0.05, 0.10, \ldots, 1.00\}$. From a simple random sample of B labels, of which b_c fall in the zone and k_c of those are errors, the exact one-sided p-value against $R(c) > \alpha$ is

$$p_c = P(X \le k_c), \qquad X \sim \mathrm{Hypergeom}(n_c,\ \lfloor \alpha n_c \rfloor + 1,\ b_c). \tag{5}$$

A rule returns a coverage $\hat{c}$ or no zone and is valid when $P(R(\hat{c}) > \alpha) \le \delta$ over the draw. The prefix rule accepts coverages from the smallest upward while $p_c \le \delta$; it is fixed-sequence testing, valid on any map whatever the shape of $R(c)$ [Angelopoulos et al., 2025]. The Bonferroni rule accepts every coverage with $p_c \le \delta/J$ over J levels; it can certify more than the prefix rule when the most confident windows hold errors. Until version 1.4.0 the package, and until 3 October 2026 the ledger, described the prefix rule as valid only where $R(c)$ does not decrease with c; that condition is not needed, and a test enumerating every error pattern of small maps confirms both rules on patterns whose risk falls as the zone grows. In the large-zone limit a zone with no errors needs about $\lceil \ln\delta / \ln(1-\alpha) \rceil$ labels (fewer on a small zone), 45 at $\alpha = 0.05$ and 255 at $\alpha = 0.009$ for the default $\delta = 0.1$; levels whose expected number of labels, cB, is below that are dropped before any label is read. The package refuses designs other than random or condition and labels that look like a review set, and certifies nothing outside the zone. A p-value within a relative 10^{-9} of its level is decided in exact rational arithmetic, so that the same labels give the same zone on every C library.

With input conditions, the L conditions holding at least the minimum at the undivided δ are fixed before any label is read (at δ/L such a condition may still certify nothing), each is certified

inside itself at δ/L, and the union of their zones errs at most α except on at most δ of samples. `certify` counts every `?` as wrong, which can only certify less. It takes no reviewer error rate: testing at $\alpha(1-E_1)$ is valid only if the miss rate holds inside every zone the test can certify, where a model's confident errors sit. In a construction from a review of 2 October 2026, misses placed in the confident half of a map, with a whole-map miss rate of exactly E_1, certified on every draw a zone that was wrong 10% of the time at $\alpha = 5\%$.

This guarantee is for one sample whose size is fixed before labelling, read once. Rerunning `certify` as labels are added runs a new test at each look, and the chance that some look certifies a wrong zone grows with the looks (Section 4.12). Since version 1.8.0 a sequential sample keeps the guarantee at every look. `sample --design sequential` draws a fixed random order of the windows, any prefix of which is a simple random sample, and fixes before any label the anchor, the smallest zone tested: by default the smallest grid level whose zone expects enough labels at a stated first budget to be certified with no error among them, at most 0.25. Each level is tested by an e-process, the mixture with equal weights over the counts $M < M_0 = \lfloor \alpha n_c \rfloor + 1$ of the likelihood ratios of sampling without replacement against M_0 [Robbins, 1970], in closed form one hypergeometric tail; it is a supermartingale under every count $M \geq M_0$, so by Ville's inequality [Ville, 1939] it ever reaches $1/\delta$ with probability at most δ, whatever decides when labelling stops. Levels are tested in a fixed sequence from the anchor outward [Bauer, 1991], valid at every look because every test in it is [Tamhane et al., 2010]; a passed level stays passed, so more labels never certify less. The closest published method, Xu et al. [2024], assumes the risk monotone in the threshold and samples with replacement; here the zones' error rates need not be monotone, which is why the sequence starts from a fixed anchor. A zone with no errors needs 71 labels at $\alpha = 0.05$ and $\delta = 0.1$, against 45 for the one-look test: the equal weights tolerate errors and pay for it on a clean zone [Waudby-Smith and Ramdas, 2020, Stark, 2023].

2.5 Which of two maps is more accurate

Where two maps of the same windows give the same class, both are right or both are wrong, so their accuracies differ only through the windows where they differ. Let N be the windows both maps predict, excluding windows split evenly between two classes when either map is a class map (with scores on both sides a tie goes to the more confident pixels), D the windows among them where the maps differ, and K_a and K_b the differing windows where map a, and where map b, matches the truth. Then

$$\Delta = \mathrm{acc}_a - \mathrm{acc}_b = \frac{K_a - K_b}{N}. \tag{6}$$

`sample a --other b` draws a simple random sample of n of the D differing windows, and the reviewer writes the class seen in each, or `?`. With a and b the sampled windows where each map is right and u the unjudged ones, each map's share of the differing windows gets its exact interval at $1 - 0.05/2 = 97.5\%$, from $\ell_a = L(a)$ and $h_a = U(a+u)$ and likewise for b, and

$$\Delta \in \left[(\ell_a - h_b)\frac{D}{N},\ (h_a - \ell_b)\frac{D}{N} \right] \tag{7}$$

with probability at least 95% by the union bound, however the `?` windows fall. The package names a map only when the interval excludes 0. It does not give either map's accuracy, which needs a sample of all windows, it assumes the labels are right, and `certify` refuses this sample. The interval is conservative; an exact interval for the difference could be narrower.

2.6 Interfaces

The command `oe-inferencex` has ten subcommands: `demo` (a shipped Dynamic World tile), `assess` (review sets, rasters and cues), `from-olmoearth` (an OlmoEarth model's output read

into the rasters `assess` takes), `compare` (disagreement and, with labels and dates, the graded block), `plan` (how many labels a question needs, new in 1.8.0), `sample` (the windows to label), `estimate` (the error rate, per-class accuracy with `--per-class`, or the accuracy difference of two maps), `certify` (the certified zone and its mask), `decide` and `mcp`. `decide` (new in 1.7.0) reads a result that `estimate`, `certify` or `compare` wrote and answers set questions with one answer from a fixed set (whether the error rate is below a threshold, a class's user's or producer's accuracy above one, which map is more accurate, the certified share), each with its level, its evidence and a sentence that holds the limit; it computes nothing new, and an interval across the threshold gives *undetermined*, not a no. Since version 1.7.0, `assess` and `sample` also take a published product as it ships, a class map with its per-pixel confidence band (`--confidence`, with `--confidence-range` for the band values that are confidences), and `estimate`, `certify` and `decide` report how many classified pixels a band's codes leave out of the population. `plan` reads only counts from the maps and gives, for each budget on a ladder, the probability over the reviewer's random draw that the package's own procedure answers: an error-rate interval no wider than a stated width at a stated rate, labels drawn where two maps differ that name the more accurate one at a stated difference, or a zone of a stated coverage certified at α (exact for the Bonferroni rule, simulated for the prefix rule on the worst map whose more confident zones are no worse); every probability is reached again in its tests by enumeration or by simulating the estimators. On a map of 10^9 windows or more, from which numpy's samplers do not draw without replacement, the prefix rule's simulation draws its labels with replacement; that moves each probability by at most $2Jn/N$ for n labels, N windows and the grid's J levels, a bound subtracted from each budget's lower end and stated in the result, and below 10^9 windows the draws are unchanged.

`from-olmoearth` reads the rslearn dataset that an OlmoEarth model's prediction writes, whether by `rslearn model predict` or by olmoearth_run, which leaves the same layout: one directory per window with its projection, bounds and completed output layers. A segmentation output becomes `scores_<EPSG>.tif`, the class probabilities of every window on one grid per CRS with NaN where no window wrote, and the command prints the `assess` command to run next. Ranking needs probabilities, which rslearn's segmentation task writes only with `output_probs` on (and, for an integer output layer, a float32 writer layer); a one-band class map is refused with that fix, as are probabilities that do not sum to 1 (a scaling or temperature applied before writing). A per-window classification output is read into one probability vector per window, checked against the class the task wrote. With `--conditions` it builds the input-condition layer of Section 2.1 from the window's own inputs: per pixel, the share of the run's timesteps a scene covered and, where every timestep holds Sentinel-2's scene classification band, the share under cloud. OlmoEarth masks a timestep missing for a whole window but reads an uncovered or cloudy pixel inside a present one as valid input, and the layer records those conditions; it does not see haze, smoke or snow the scene classification does not flag. The reader needs no torch and no rslearn; it was tested on synthetic rslearn datasets built to rslearn's source, and exp98 and exp99 are its runs on real outputs (Section 4.5).

Since version 1.4.0, `oe-inferencex mcp` serves the commands as tools (`guide`, `assess`, `compare`, `sample`, `estimate`, `certify`, from 1.7.0 `decide` and from 1.8.0 `plan`) to an agent over the Model Context Protocol on local standard input and output, with nothing hosted. Each tool runs the command line's own code in process, so a tool and its command give the same numbers and refuse the same inputs, and returns compact JSON with the files written, the summary numbers, and three plain texts: the conclusion with its main limit, the limits, and the next step. The server's instructions state six hard rules: a review set is not a sample, and `estimate` and `certify` refuse it; `certify` needs a random, sequential or condition sample; `compare` cannot say which map is right without labels; labels are assumed right, and only `estimate` widens its interval for a reviewer who errs, at rates the user states; ranking needs the scores, not only a class map, or a product's class map with its confidence band; and an interval or a certificate holds for one sample

whose size was fixed before labelling, read once with the rule, α and δ chosen before the labels (from 1.7.1; a sequential sample's certificate holds at every look), so that a second sample, or a second run with another rule, α or δ, is a second test whose error levels add.

3 Evaluation protocol

3.1 Units, statistics and controls

The graded unit is the window on segmentation maps and the sample on classification tasks. With n units and k errors, the selective risk at a coverage is the error rate among the least suspect units kept; AURC is its mean over the n coverage levels, in expectation over orderings of tied units. The perfect ranking has $\mathrm{AURC_o} = n^{-1} \sum_{i=1}^{n} \max(0, i - (n - k))/i$, the excess AURC is $\mathrm{AURC} - \mathrm{AURC_o}$ [Geifman et al., 2019], and since a random ranking scores $e = k/n$, a reading takes the share $1 - \text{E-AURC}/(e - \mathrm{AURC_o})$ of the ranking headroom. The capture at budget b is the review set's share of all errors, reported beside its ceiling $\min(1, b/e)$. For a fixed error set AUGRC [Traub et al., 2024] is a decreasing affine function of the failure AUROC. Readings are compared by one-sided exact sign tests over groups (tiles, events, regions, tasks), which are not all independent: 15 of the 24 suite tasks share five sources, and GEOID-Flood's areas cluster into 9 and 10 activations. Tests are design-weighted [Olofsson et al., 2014] where the reference is a probability sample.

A candidate signal is supported only if it beats both the model's confidence and a control that does not use the model's scores, and the margin must beat the control. On imagery the controls are pixel statistics such as a water index; on the published suite, whose imagery is not released, they are a sample's embedding distance from the training mean and the rarity of its predicted class, which reads the probe's decision and so is not wholly free of the model. Section 4.2 shows that both suite controls are near chance, so a win over them shows little beyond a win over a random order.

3.2 Kinds of reference

Table 1 lists the labelled testbeds. The OlmoEarth paper's embedding suite [Herzog et al., 2025] adds 24 tasks with splits fixed by the model authors; as every unit is labelled, estimators are graded on it over 2,000 simulated draws of 300 labels.

Origin of the reference	Testbeds
people interpreting the imagery	Sen1Floods11 [Bonafilia et al., 2020], GEOID-Flood (Copernicus EMS) [Chiriaco et al., 2026], WorldFloods v2
another model's output	DFC2020 [Yokoya et al., 2020], an iterated random forest
experts annotating a product	Dynamic World [Brown et al., 2022], 409 tiles
surveyors in the field	LUCAS Copernicus 2022, the module described for 2018 by d'Andrimont et al. [2021]
farmers' declarations	EuroCrops [Schneider et al., 2023], labelled in both years
the model authors' validation sets	Ai2's fine-tuned FT-AWF and Forest Loss Driver models
analysts interpreting a simple random sample	LCMAP Reference Data Product v1.2 [Pengra et al., 2020], 25,000 plots

Table 1: Labelled testbeds by origin of the reference.

3.3 Preregistration, the claim ledger and corrections

Each experiment states its predictions and falsification conditions before it runs, with dated amendments; results read afterwards are marked as not preregistered here and in the ledger. The repository holds 92 experiment scripts named `expNN_*.py`, covering 83 experiment numbers from exp01 to exp100. The ledger `docs/claims.yaml` holds 228 claims, each with a statement, status, artifacts and an executable check (two rest on no committed artifact and say so); 75 are supported, 78 measured, 32 mixed, 41 rejected and 2 superseded, and `scripts/claims.py stale` flags only the two superseded ones. After an audit on 23 September 2026 found that 138 of the then 167 checks read a result back from its own artifact, second routes were added: 91 of the 228 claims name a `crosscheck` test that reaches the number by another route, and every claim from exp79 on must name one.

Corrected results appear in their current form. exp78's cluster-corrected arm is now the ratio estimator, which raises coverage to 0.913 to 0.932 on the four tasks with full tiles but not on Sen1Floods11 (0.824) or MADOS (0.598), and the earlier statement to the contrary was withdrawn. The finite-population Wilson interval is now the score inversion at the effective size $n(N-1)/(N-n)$, after the earlier form excluded a true rate of 0.010 whenever a 100-window class's single error was sampled in 30 labels. exp70's third prediction, reported at $p = 6.9 \times 10^{-15}$ from 144 task pairs treated as independent, has $p = 0.0054$ by a rank-sum test over tasks and still holds. exp54's cross-encoder ϕ block was withdrawn after a row-alignment defect was found while its own check passed.

A red-team review on 30 September 2026 changed how the suite result is read. Both suite controls are near chance, so the headline now leads with the ranking headroom and the capture at a 10% budget rather than with the win over the controls [suite-controls-are-near-chance]; six of the seven segmentation outcomes had already been scored on the same windows before exp70's preregistration, which was not disclosed when exp70 was recorded [suite-six-segmentation-outcomes-were-on-record]; and exp88's confident errors were attributed to a probe read on an input combination it was not trained on rather than to the missing sensor [exp88-matched-head-ranks-normally]. On 3 October 2026 a review of exp90 replaced a whole-map baseline that understated what random labels allow by about half (Section 4.9).

4 Results

4.1 Ranking on the labelled testbeds

The model's own confidence is the best single label-free error ranker on the AWF points, the fine-tuned model and the multi-region Sen1Floods11 test split, for Sentinel-2 heads under both backbones and both window designs, leading or tying [confidence-best-single-signal]. On the median of 45 GEOID-Flood areas of interest, the 5% of windows it flags hold 88% of the permanent-water and 64% of the post-event water errors, 17.5 and 12.8 times a random 5% [geoid-capture-effect-size]. On Sen1Floods11 Bolivia, one flood event, the NDWI control ranks the errors as well as confidence under OlmoEarth v1 and better under v1.2 Base [bolivia-ndwi-exception]; under a Sentinel-1 probe NDWI ranks them better on the whole multi-region test split [s1-probe-ndwi-flip]; over the GEOID-Flood areas the exception rate is one activation in nine [geoid-exception-rate]; and after fine-tuning, the model's confidence beats the NDWI control on Bolivia under both backbones [finetune-dissolves-bolivia-exception].

4.2 The published embedding suite

Of 25 suite tasks, 24 define a margin. Read through linear probes on OlmoEarth Base's frozen embeddings, the margin closes a median 0.68 of the gap between a random and a perfect order [margin-takes-two-thirds-of-the-ranking-headroom], and a review of the least confident 10% of units finds a median 0.214 of the errors, against 0.10 for a random 10% and a median 0.410 for a perfect order; it finds more than 10% on every task, the lowest 0.115 on Nandi Sentinel-1 [suite-review-at-ten-percent]. On all 24 tasks and all 14 distinct sources the margin also beats the better of the two no-model controls, with leads of 0.0157 to 0.2278 of excess AURC over 6,435,473 units (sign test over tasks $p = 6 \times 10^{-8}$, over sources $p = 6.1 \times 10^{-5}$) [suite-margin-wins-every-task], including all six GEO-Bench tasks [Lacoste et al., 2023] [suite-holds-in-both-families]. Those controls close a median 0.094 of the gap and do no better than a random order on 9 of the 24 tasks, so this win mostly shows that the margin beats a random order [suite-controls-are-near-chance]. On the 17 tasks whose outcome no earlier artifact held, the margin wins 17 of 17 [suite-six-segmentation-outcomes-were-on-record]. Under ten probe seeds the lead stays positive on all 24, the smallest being +0.0017 on CropHarvest Togo Sentinel-1 [exp79-base-margin-wins-under-every-seed]. The margin beats a five-seed ensemble [Lakshminarayanan et al., 2017] on 22 of 24 tasks and the nearest-neighbour [Sun et al., 2022] and Mahalanobis [Lee et al., 2018] distances on 24 of 24 [suite-margin-beats-the-strong-alternatives]; only the ensemble beats it, on EuroSAT and AWF Landsat [suite-alternatives-clear-the-control-and-still-lose].

Across the sixteen encoders of the suite (OlmoEarth and eight other families), the margin beats a random order in every one of 3,560 cells of encoder, task and probe seed, the lowest AUROC being 0.503 for Clay Large on Nandi Sentinel-1, close to chance [exp79-margin-beats-random-everywhere]. At seed 0 it beats the best control on 322 of the other fifteen encoders' 332 (encoder, task) pairs [suite-holds-under-every-encoder]; under each of ten seeds it does so on at least 75% of the tasks of all sixteen encoders, the lowest share 0.875 [exp79-headline-holds-under-every-seed-on-every-encoder]. Its 122 failures over the 150 encoder-seeds of the fifteen lie on the three 306-unit CropHarvest Togo tasks or on Nandi Sentinel-1 at a probe accuracy of at most 0.40 [exp84-losses-are-small-or-badly-fitted-tasks], and the per-encoder sign test survives a Holm correction at every seed [exp84-bar-survives-holm]. Across encoders the median share of the headroom runs from 0.551 to 0.693, and its standard deviation between tasks is 2.8 times that between encoders within a task (descriptive, not preregistered) [the-ranking-ceiling-is-the-tasks-not-the-encoder]. The best encoder is OlmoEarth Large or Base under every seed, so the record names a set of leading encoders [exp79-encoder-ordering-stable-as-a-set]. One minus the top probability ranks errors better than the margin on 14 of 16 multi-class tasks, by a median 0.0006 of excess AURC [top1-beats-the-margin-on-multiclass], and by AUROC, hence by AUGRC, the margin beats the best control on 23 of 24 tasks [augrc-leaves-the-suite-result-standing].

4.3 References produced independently of this project

On Dynamic World's 409 expert-annotated tiles, the margin of the product's published probabilities ranks its errors better than a control that uses neither imagery nor probabilities, excess AURC 0.0637 against 0.1120 [dw-margin-ranks-a-production-model]; one minus the top-1 probability ties on 23.4% of windows and the margin on 3.6% [dw-margin-beats-naive-confidence]. The published top-1 probability understates the product's accuracy by about 0.20 (0.539 against 0.738), so the same numbers order a review well and threshold badly [dw-published-probabilities-are-underconfident]. On LUCAS, where a surveyor observed the point, the margin's design-weighted excess AURC is 0.1468 against 0.2421 for the best inference-time control [lucas-ranking-survives-ground-observation]. On EuroCrops it beats the best control in all three countries and on all but four of 147 grid cells [eurocrops-ranking-holds-on-declarations], and on DFC2020 it beats the NDWI control on all three sensor arms [dfc2020-margin-beats-pixel-control]. Unlike the suite's, the best no-model control on Dynamic World, EuroCrops and LUCAS finds errors above chance, for instance 0.217 of Dynamic World's errors in a 10% review, so these wins are over informative controls (read after the red team, not

preregistered) [external-references-carry-informative-controls].

4.4 Fine-tuned models

A replica of Ai2's fine-tuned AWF model reaches 0.881 accuracy against the reported 0.895 on its 344 validation points (41 errors); confidence and tiling instability rank its errors best and are tied [fine-tuned-model-audit], and the logit margin catches 22%, 39% and 63% of the errors at review budgets of 5%, 10% and 20% [fine-tuned-capture-at-budgets]. exp89 added Ai2's Forest Loss Driver model (109 windows, 25 errors) as a report-only arm, its validation set being too small to grade: the top-1 confidence ranks the errors of the two models with AUROC 0.867 and 0.808, and a 10% review holds 41.5% and 32.0% of the errors against 19.5% and 18.3% for the best no-encoder control, a lead whose cluster-bootstrap 95% interval includes zero on both [exp89-finetuned-report-only].

4.5 A deployed OlmoEarth model

exp98 and exp99 ran FT-AWF through Ai2's deployment pipeline instead of the evaluation's crops: Ai2's checkpoint, the olmoearth_run configuration published with the v1 models (olmoearth_projects at f3c9b0c8) and Planetary Computer's Sentinel-2 scenes, at 10 m, with two changes so the map could be graded, the softmax written instead of the argmax and a sidecar reading the scene classification band. Ai2's lock at that commit installs olmoearth-runner 0.1.12 with rslearn 0.0.23; the record chose 0.1.14 with rslearn 0.0.27, which has `output_probs`, and patched rslearn's `get_item_by_name`, a bug that belongs to that choice. The current way to run these models is rslearn's own prediction, so "deployment" here means that v1-era pipeline as re-run, and neither map is one Ai2 published. Both were preregistered and frozen before any prediction was read at a label, read with `from-olmoearth`, and audited in three lenses before recording, which recomputed every number it could and changed none.

exp98 re-ran the configuration over its own 2023 request geometry (270 windows, all 12 timesteps) and graded the map at the 259 of Ai2's 344 validation points that fall on it (20 annotation tasks, 33 errors). It is 87.3% accurate there, against 89.6% for exp89's replica of the evaluation on the same points; the 90% task-cluster interval of the difference, -5.1 to $+0.8$ points, reaches 0.15 points past the preregistered ± 5-point margin, so reproduction within 5 points was not shown, and with the interval containing 0 (sign test $p = 0.26$) no difference was shown either [exp98-equivalence-within-5-points-not-shown]. The verdict sits at the margin: over 200 other bootstrap seeds the larger end ran from 4.8 to 5.3 points, and the prediction would have held at 40 of them. The re-run gives the replica's class at 91.9% of the points, its own top-1 confidence ranks its errors with AUROC 0.849 (the replica's 0.887; the 95% interval of the difference -0.090 to $+0.026$), and the least confident 10% of the points hold 42.4% of its errors [exp98-deployment-gives-the-evaluations-classes-and-ranking]. The points sit in the confident part of the map: on average a point is more confident than 64.7% of the map's pixels (one-sided 95% lower bound 61.4%, against 50% if they were placed as the map's pixels are). The design does not say why, and it gives no map accuracy; only under the unchecked assumption that points and pixels of equal confidence are equally accurate would the points' error rate, reweighted to the map's confidence fifths, read 21%, a model-based reading with no interval resting on 22 points in the least confident fifth (12 errors) [exp98-validation-points-sit-in-the-confident-part]. Ai2's points were placed by experts, not drawn at random, so exp98 gives no whole-map error rate, per-class user's or producer's accuracy or certified zone; every figure describes those points.

exp99 held the model and configuration fixed and moved the year to 2017, to grade the map at the one probability sample near the AWF area, the East Africa TimeSync plots (a simple random sample within each country), through a crosswalk from their legend to AWF's, STRICT or LENIENT. A rule fixed before any map took the 309 plots within 100 km of Ai2's request geometry; 42 of the 313 windows received no 2017 Sentinel-2 item group at all, and the 38 plots

that lie only in them have no input, so 271 plots were graded. The 38 are not missing at random, as the preregistration had assumed: 20 of the 47 plots inside Ai2's geometry lack input against 18 of the 262 outside (Fisher $p = 4 \times 10^{-9}$), and cropland and open grassland lack it more often than wooded grassland. Every rate and AUROC below therefore describes the plots with 2017 input, which under-represent Ai2's own area and cropland, not the 100 km region. The design-weighted AUROC of the confidence for the STRICT errors is 0.638 (90% bootstrap within countries 0.579 to 0.696), below the preregistered 0.70; LENIENT's 0.678 also misses on the point value, but its interval, 0.599 to 0.756, contains 0.70 [exp99-confidence-misses-the-ranking-bar-under-both-rules]. The frozen rule labels both misses "transfer is the likelier reading"; that is the rule's label, not a finding, since the run does not separate the year (the model was fine-tuned on 2023 imagery), the place, the grass/shrub legend boundary and the move from expert-placed points to a random sample, and the plots sit in their windows as random pixels do (mean percentile 0.48, against 0.647 for Ai2's points in exp98). The least confident 10% of the plots by design weight close 0.434 of the gap between a random and a perfect review (bar 0.25), so that prediction holds on the point value, but its interval, 0.10 to 0.65, includes the bar, the 10% is 31 plots, and the figure is specific to the budget (0.169 at 20%) [exp99-least-confident-tenth-closes-the-gap-at-the-point-value]. Report-only, with Kenya and Tanzania as strata, the error rate of the plots with 2017 input is 35.5% under STRICT (the package's exact interval, at least 95%: 26.0% to 45.8%) and 13.4% under LENIENT (7.5% to 21.9%), so whether the map is worse than Ai2's 10.5% error by more than 5 points is not determined; the rules differ mostly at the grass/shrub boundary, for which AWF publishes no threshold. With the 38 plots without input filled all right or all wrong, the rate over all 309 lies between 31.2% and 43.4% under STRICT and between 11.8% and 24.1% under LENIENT [exp99-error-rate-of-plots-with-2017-input]. These are the record's first design-based error rates for an OlmoEarth deployment, at the plots with 2017 input, for one model and one year, with labels taken as right.

4.6 Input conditions

exp88 read the PASTIS probe trained on OlmoEarth Base's Sentinel-1 plus Sentinel-2 embeddings, without refitting, on the Sentinel-1-only embeddings of the same 458,638 windows, the extreme case of a cloud. It errs on 73.6% of them against 19.5% with full input, and 59.8% of its errors carry a margin at or above the median margin of the windows full input gets right, against 6.0% [missing-optical-errors-are-confident]. A probe trained on the Sentinel-1 embeddings alone errs on 28.4%, only 3.9% of its errors reach that margin, and its margin ranks errors with an AUROC of 0.79, against 0.59 for the mismatched probe, so the confident errors come from reading a probe on an input combination it was not trained on; this reading was not preregistered, and the two probes were trained at different learning rates. On OlmoEarth Large the mismatched probe's confident share is 12.8% [exp88-matched-head-ranks-normally]. The rise does not replicate on CropHarvest China 6 with the optical input missing, and appears there with radar missing and on Togo 12 with the optical input missing [missing-modality-confident-errors-follow-the-shift-not-the-sensor]. Inside the degraded condition the margin still beats the best no-model control [margin-still-ranks-inside-the-cloudy-part].

On a PASTIS map with half its tiles read without the optical input, the cloudy part errs on 74.1% of windows and the clear part on 19.7%. A random sample of 300 windows estimates 46.9% on average and its interval covers the cloudy part's rate on none of 2,000 draws; split by condition, each part's interval covers its own rate on 95.6% and 95.7% of draws. A 5% review ordered by margin across the whole map holds 4.3% of the cloudy part's errors and 18.3% of the clear part's, although the cloudy part carries 79% of all errors [pooled-error-rate-misstates-the-cloudy-part]. These results are the reason for the condition layer of Section 2.1; no real cloud has been tested.

4.7 Explanation cues and review order

On Bolivia hand labels, 75% of error windows lie on a prediction boundary against 21% of correct windows [errors-sit-on-boundaries], and 95% of error windows carry at least one measured cue [explanation-cues-cover-errors]. The boundary-first order captures more errors than the confidence order at 5% and 10% there and not at 20% [boundary-first-review-order]. Elsewhere it wins at 5% on a minority of GEOID-Flood events [geoid-boundary-first-minority], and on the suite's Sen1Floods11 split it captures fewer errors at 10% and 20% (0.415 against 0.459; 0.650 against 0.695) [boundary-first-loses-on-the-suite-flood-split]. The package's default is confidence order.

exp82 verified the cues on the seven segmentation tasks. The boundary cue's enrichment runs from 1.24 (m-cashew-plant, 76% of windows on a boundary) to 8.13 (MADOS, 10%), with a rank correlation of -0.26 with the number of classes and 1.00 with the ceiling $(1-\theta)/(p-\theta)$ for boundary prevalence p and error rate θ; the error rate inside the boundary set is at least 2.1 times that outside on every task [boundary-cue-enrichment-is-fragmentation-not-class-count]. The Bolivia value formerly quoted on every map, 3.5, lies outside every task's 95% interval [cue-library-value-outside-every-task-interval]; windows with both the boundary and low-confidence cues have a higher error rate than windows with either [cue-conjunction-is-purer-than-either-cue]; and the cue weakens as windows grow [boundary-cue-weakens-with-the-window-grain], to an inside-to-outside error-rate ratio of 0.81 on AnySat's coarser m-cashew-plant grid [cue-verification-holds-on-other-encoders].

4.8 Comparison of two maps without labels

For every pair of inferences compared (crop offsets, backbones, sensors, encoders, frozen against fine-tuned), the differing windows carry the boundary cue 3.3 to 7.1 times as often as the agreeing ones on Sen1Floods11 and 21 times on the median GEOID-Flood event [atlas-disagreement-is-boundary-located], and the sensor and backbone differences occupy different windows (ϕ 0.27 and 0.18) [atlas-different-differences]. On DFC2020 the two sensors differ on 44.27% of windows, 33 times the probe-seed rate [dfc2020-sensor-difference-dominates-land-cover].

No label-free rule in the record decides which map is right. Believing the side with the larger margin is right on 51% to 70% of differing windows, above a coin flip on every pair and below the better side wherever one is known [tool-vs-diff-resolution]. A side rule fitted on labels and cross-fitted by tile beats it by 8 to 24 points on all eight Sen1Floods11 pairs [calibrate-side-rule-held-out], but a rule fitted on frozen pairs and forced onto the frozen-versus-fine-tuned pair is right on 43.8% and 49.2%, below the raw rule's 61.1% and 55.4% [calibrate-family-lock]. A label-fitted fusion of five readings lowers confidence's held-out excess AURC from 0.0105 to 0.0084 on Bolivia and from 0.0098 to 0.0069 on the test split [calibrate-ranker-fusion-beats-confidence]. The package therefore reports both sides and locks each fitted rule to its model family.

Across dates a difference can be change on the ground. On GEOID-Flood a same-period cross-sensor difference is flooded by the label on 0.4% of its windows, against 19.5% for a pair mixing sensor and date [two-periods-sensor-axis-isolated]. On LUCAS, two acquisitions a median 118 days apart change 31.2% of decisions against a reseed floor of 0.30% [lucas-two-dates-are-phenology]. On EuroCrops, which labels both years, decisions change on 0.060 to 0.366 of windows whose declared crop did not change, one to two orders of magnitude above the reseed rate [eurocrops-the-labelled-floor-for-a-two-date-difference], and on 0.051 to 0.367 of differing windows the model is right in both years [eurocrops-a-difference-can-be-the-model-tracking-the-ground].

4.9 Which of two maps is more accurate, with few labels

exp90 checked the design of Section 2.5 on every pair of encoders run on the same units of each of the 24 suite tasks, 2,514 pairs with the full truth and 400 draws per pair and budget. It was not preregistered, and the maps are the record's linear probes, not map products. The paired maps

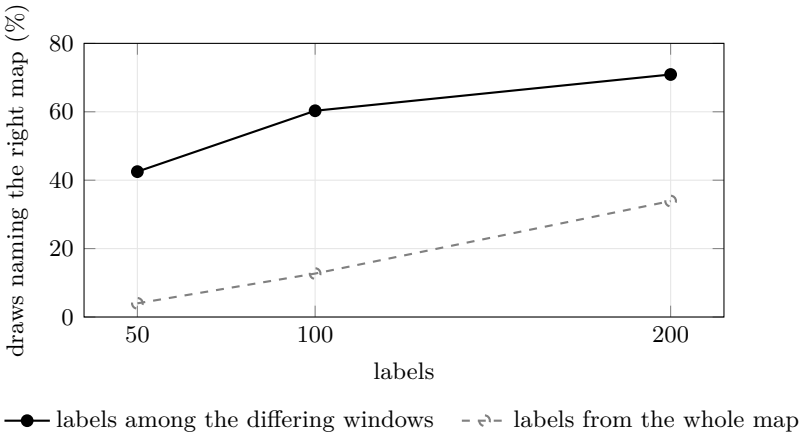


Figure 4: exp90, means over 2,514 pairs of encoder maps: how often the 95% interval on the accuracy difference excluded 0 on the right side. Both curves use the same interval; the whole-map labels are analysed on the differing windows they hold. At 200 labels the whole-map draw labels the entire map on the 268 pairs of the three 200-unit AWF tasks. [exp90-which-map-few-labels]

differed on a median 20.7% of windows, and the median absolute true accuracy difference was 3.4 points. Labelling a random sample of the differing windows named the right map on 42.5%, 60.3% and 70.9% of draws at 50, 100 and 200 labels, against 4.0%, 12.7% and 33.9% for the same number of windows drawn from the whole map and analysed on the differing windows they held (Figure 4). Pairs that differ on no more windows than the budget are labelled in full; on the other pairs the rates were 36.5%, 42.8% and 56.7% against 4.2%, 11.3% and 23.1%. Where the true difference was 5 to 10 points, 100 labels named the right map on 80.7% of draws, and above 10 points on 96.4%. The lowest coverage over the pairs was 95.5% at each budget, a Monte Carlo share with a standard error of about one point, with medians of 99.25% to 99.5%, and the most draws of any pair naming the wrong map were 3.75%, 4.0% and 1.75% [exp90-which-map-few-labels]. An earlier analysis of the whole-map draw, with each map's count taken over all N windows, gave 1.2%, 6.3% and 25.2% and was replaced because it understated what random labels allow.

4.10 Published products against an independent probability sample

exp92 graded two labelled routes, the error-rate interval and the comparison of two maps, and the ranking on published products; the certified zone was not graded. The reference is LCMAP's Reference Data Product v1.2 [Pengra et al., 2020]: 25,000 plots, a simple random sample of the 30 m pixels of the conterminous United States, interpreted into LCMAP's eight classes; for 2018 the maps are LCMAP Collection 1.3 primary land cover with its confidence, and Esri's 10 m Annual Land Use Land Cover v2 crosswalked to LCMAP's classes. The 24,968 plots where the reference and both maps have a class stand in for the map population, so every design is graded exactly against their truth. All four preregistered predictions held. LCMAP is 82.3% and Esri 79.6% accurate against this reference, and the maps differ on 21.6% of plots. Labels drawn among the differing plots covered the 2.7-point difference on 98.5% to 99.0% of 2,000 draws at 50, 100 and 200 labels, at about half the width of labels drawn from all plots, and never named Esri; at so small a difference they named LCMAP on 12.0% of draws at 100 labels (2.5% from all plots). Three hundred random plots gave an exact error-rate interval that covered on 95.0% and 95.65% of draws [exp92-products-labelled-routes-hold]. LCMAP's own confidence ranked its errors with an AUROC of 0.753 (bootstrap interval 0.734 to 0.771), and its 10% least confident plots held 28.7% of the errors [exp92-lcmap-confidence-ranks-its-errors]. After the result, the comparison was found to need about 800 labels to name LCMAP on 95% of draws, against more than 3,200 for labels drawn from all plots. A pre-record audit found that the first extraction had left out 734

plots with Esri data; the numbers here are corrected, with the same verdicts. The reference uses LCMAP's legend, so Esri is graded through a crosswalk at a finer resolution, and the difference measures agreement with this reference rather than which product is better in general. Whether any reference plot was used to train LCMAP Collection 1.3 was not checked; the evidence is one year in one country.

Five more reference samples from four sources extend this (exp94, preregistered): NLCD's 2011 and 2016 accuracy-assessment points (stratified, the United States, 2001 to 2016), JRC's GFC2020 validation set (stratified, global forest, 2020), the East Africa TimeSync sample (random within seven countries, 2015 to 2017) and the S2GLC validation set (Europe, 2017, not a probability sample), graded against seven products: NLCD against LCMAP, GFC2020 against ESA WorldCover, CGLS-LC100 against Esri, ODSE-LULC against Esri. The points stand in for the map, as above. Labels drawn where two products differ covered the difference on 98.05% to 99.25% of draws in all eight pairs, at a third to two thirds of the width of labels from all points, and named the worse-agreeing product on at most 0.35%; on the four pairs differing by 4 to 6 points, 200 such labels named the better-agreeing one on 83% to 95% of draws, against 13% to 25% from all points. The exact error-rate interval covered on 95.05% to 96.25% of draws in all 18 cells [exp94-labelled-routes-hold-on-four-references]. LCMAP's, CGLS-LC100's and ODSE-LULC's own confidences ranked their errors, AUROC 0.720 to 0.787 in nine cells, every bootstrap lower end at least 0.704 [exp94-product-confidences-rank-their-errors]. A pre-record audit recomputed 346 quantities by independent code and changed none. Several pairs are related: LCMAP was trained on NLCD's 2001 map, and GFC2020 is built partly from WorldCover (their errors correlate at 0.51); ODSE has no class over the sea, and CGLS no probability for water or built-up land, so those points are outside the gradings.

4.11 Estimation from a labelled sample

exp78 graded the estimators on the suite's seven segmentation tasks, from an export that reproduces exp70's accuracies [exp78-export-reproduces-the-suite]. With 300 randomly sampled windows the finite-population Wilson interval, the form the package printed at the time, covers the true error rate on 0.933 to 0.961 of 2,000 draws on all seven [design-based-interval-is-honest], with a half-width of about 3 points at a low error rate and 5 at a high one; on all sixteen encoders it covers 0.933 to 0.963, and the exact interval the package now prints has an exact coverage of at least 0.9506 on the same 111 cells [exp79-estimation-findings-hold-on-every-encoder]. Taken as 18 tiles of 16 windows with the independent-sample formula, the interval covers on 0.506 to 0.777 of draws on six of seven tasks; the ratio estimator covers 0.913 to 0.932 on the four tasks with full tiles, 0.824 on Sen1Floods11 and 0.598 on MADOS, whose tiles hold 1 to 400 windows [tile-sampling-breaks-the-naive-interval]. The tile design as the package ships it, checked without preregistration, covers 94.5% to 95.4% on five tasks and falls short on Sen1Floods11 (84.3%) and MADOS (68.5%), the two tasks where a tenth of the tiles hold most of the errors [shipped-tile-design-coverage]. The confidence design narrows the interval to a median 0.80 of the random width with coverage intact, but its clause requiring a ratio above 0.94 at over 30% error failed at 0.9361 [confidence-helps-most-where-the-map-is-good]; it saves up to 2.51 times the labels, above the preregistered 1.8 [confidence-saves-a-quarter-to-a-half-of-the-labels]. Labelling the 5% review set and dividing gives 1.8 to 5.8 times the true rate on the seven tasks (from exp70's capture at 5%), the case the review-set check refuses.

Each condition's exact interval, applied to a random sample of 300 windows split by condition, was graded on exp88's half-cloudy map above, covering 95.6% and 95.7% [pooled-error-rate-misstates-the-cloudy-part]; the condition design's equal allocation and its whole-map union-bound interval were not. The whole-map union-bound interval, the `?` interval, the reviewer-error interval, the per-condition certificate and the interval on the difference of two maps are checked by exact enumeration on populations of 10 to 100 windows in the test suite, where each keeps its stated guarantee against every error pattern (the per-condition certificate against five patterns per condition on

every draw), every adversarial choice of unjudged windows and every reviewer within the stated rates. Apart from the per-condition intervals above and exp90, none of them has been graded on a real map in a recorded experiment.

For per-class accuracy (exp81), the Wald interval of the map-accuracy literature leaves 107 of 522 cells below 0.93 coverage, the worst at 0.019, as it collapses to a point when a class shows no sampled error; the package's forms leave 14 of 628, none below 0.879, and no random-design user's-accuracy cell below 0.947 [per-class-intervals-wald-fails-wilson-nearly-holds]. On the nine rare-error cells the few-errors warning fires on 1,081 of the 1,351 draws whose interval misses [rare-error-warning-marks-the-misses], and the confidence design's per-class user's-accuracy intervals at 300 labels are narrower than a random sample's on 19 of 21 tasks [confidence-design-serves-the-classes-too]. exp85 measured a model-assisted difference estimator with a tuned coefficient (PPI++; Angelopoulos et al., 2023b). Under a random sample it covers 0.94 to 0.95 at 0.840 to 0.930 of the Wald width [model-assisted-arm-is-honest-and-never-wider]; under the confidence design its stratified form [Fisch et al., 2024] saves 3.5% to 5.4% of labels at 0.4 to 1.0 points of coverage, and by the preregistered rule it was not adopted [model-assisted-gain-beyond-the-design-is-two-percent].

4.12 The certified zone

On OlmoEarth Base's 24 tasks at 100, 300 and 1,000 random labels with $\delta = 0.1$ (exp80), the prefix zone exceeded α on at most 0.080 of 2,000 draws and the Bonferroni zone on at most 0.0045, on all 112 cells (110 with a testable level), against a bound of 0.120; the plug-in zone, the largest with a sample error rate of at most α, exceeded it on up to 0.557 [trust-zone-guarantee-holds]. At 300 labels and α half the error rate, the plug-in exceeds α on more than δ of draws on 16 of 21 tasks [plugin-zone-violates-on-most-tasks], and the prefix rule certifies a zone on most draws on 14 of the 21 tasks, its median certified coverage 0.50 against an oracle median of 0.60 [trust-zone-coverage-at-300-labels]. The guarantee held on four other encoders too [trust-zone-guarantee-holds-on-other-encoders]. On a published product's own confidence (exp93: LCMAP Collection 1.3's `lcpconf`, against the 4,796 plots of exp92's confidence subset standing in for the map, 17.7% wrong), the prefix zone exceeded α on at most 6.1% of draws and the Bonferroni zone on at most 0.2% in all 24 rule-grid-cells, and the plug-in on 46.6% at 300 labels and α half the error rate; there the prefix rule returned no zone on 87.45% of draws, as summed exactly before the run: the first level tested, a tenth of the plots (those at 100 and a random part of the tie at 99), is 6.0% wrong against $\alpha = 8.86\%$ and holds about 30 labels [exp93-zone-guarantee-holds-on-a-product]. LCMAP's confidence is coarse, 55% of the plots at 99. At $\alpha = 8.86\%$ the prefix rule returned a zone more often on the product's own thresholds, `lcpconf` ≥ 99 (57.6% of those plots, 7.2% wrong) or more on 25% of draws at 300 labels and 60% at 1,000, than on the package's grid (13% and 39%); Bonferroni, and the prefix rule at $\alpha = 5\%$, went the other way (descriptive) [exp93-grid-choice-changes-what-is-certified]. Plots whose `lcpconf` is a provenance code (4%, 29% wrong) are outside the zone's population and must be masked before certifying. On three products' confidences against exp94's references, at 300 labels and α half the error rate (5.2% to 20.9%), the prefix zone exceeded α on at most 2.8% of draws and the Bonferroni zone on at most 0.4%, the plug-in on 39% to 53%, and the prefix rule returned no zone on 73% to 97% of draws [exp94-zone-guarantee-holds-on-three-products]. The mean top-1 probability exceeds accuracy by a median 0.061 [mean-confidence-overstates-accuracy], which is why the zone is calibrated on labels.

The guarantee is per budget, not per look. On nested samples, a larger budget adding labels to a smaller one, the prefix rule's mean certified coverage fell by more than 0.05 of the map when labels were added on 14 of 34 maps (exp95, preregistered; a level enters at the budget where it expects the minimum number of labels and must then be passed with about that many), and a cut that tests a level later halved those falls but certified less at about 100 labels on 24 of the 34, so the standard cut stayed [exp95-standard-cut-falls-when-labels-are-added]; a second cut designed after exp95 passed at its preregistered seed on 332 other maps and not across seeds, and is an option, not the

default [exp96-ramp-passed-at-its-seed-and-not-across-seeds]. Certifying at every budget of a doubling ladder on six of those maps, where one budget's zone was wrong more than α on at most 12.25% of draws, gave a wrong zone at some look on 21% to 30% [certify-guarantee-is-per-budget-not-per-look]. exp97 (preregistered) graded the sequential certificate on the 366 maps: over 3,122 violation rates the largest was 0.1175 at $\delta = 0.1$ (bound 0.175, none above three standard errors), and its zone never shrank [exp97-sequential-certificate-holds-at-every-look], where `certify` rerun after every 5% more labels, a median 77 looks, certified a wrong zone at some look more than 17.5% of the time on 132 maps, at most 0.435 [exp97-rerunning-certify-breaks-its-guarantee]. It costs labels: with the anchor of a first budget of 300, a median 1.71 times the one-look rule's labels to reach half the largest mean certified share that rule reaches (213 maps), and over budgets 50 to 3,000 a median mean certified share of 0.146 against 0.222 on the same 279 maps [exp97-sequential-certificate-costs-labels]. The default stays the one-look certificate; the sequential one is for a reviewer who will look more than once.

4.13 Use by a language-model agent

exp64 preregistered whether the package helps an agent, on forty cards with hidden labels, using `nvidia/Qwen3.8-27B-NVFP4`. With the package as tools, the model reproduced the review set, grounded 99.4% of its numbers and declined to choose a side on every comparison card [agent-benchmark-tool-arm-reproduces-the-package]. Two predictions failed. A numpy sandbox captured 0.904 of the package's errors [agent-benchmark-sandbox-rediscovers-the-ranking] and grounded 94.2% of its numbers against 99.4%, a gap below the preregistered 0.2 once a grading that read only JSON fields was withdrawn [agent-benchmark-grounding-not-decisive]. The tool arm's decline held as graded, but an arm with only the card's text, no data and no package, also declined on 30 of 30 runs, so the benchmark cannot show that the package causes the decline [agent-benchmark-decline-holds]. At 7B all three predictions held [agent-benchmark-package-is-a-floor-at-7b], and the shipped agent called the review-set tool on 17 of 40 runs against 40 of 40 at 27B [agent-benchmark-7b-agent-does-not-find-the-tool].

Two later agent results carry no ledger claim and are stated with their limits. In the OlmoEarth Agent trial exp86, material false statements per sentence fell from 7.5% to 2.2% over development rounds 7 to 10, on the eight briefs the fixes were built against; the trial passed only under an instrument amended fourteen times after results were seen, and under the preregistered instrument its answers fail three of its predictions. The held-out trial exp87 has been run and audited but not graded. Both trials' files are archived at tag v1.9.0 (Section 8). In a pilot of 2 October 2026, agents answered five questions through the MCP server on the outputs of the two fine-tuned models of Section 4.4: the tools' numbers were right, a strong model's replies kept the numbers and their limits, and a small model's replies failed on all five, mostly by dropping limits; after the server's texts put each tool's main limit inside its conclusion, two of the small model's five replies passed. The pilot was not preregistered and left no committed artifact; `estimate` and `certify` refuse a review set, so a weak model's advice to label one cannot become a wrong number, though the reply can still mislead.

4.14 Conditions for reading a ranking

On LUCAS, a probe that memorised its fit set placed the margin last of four model signals and a regularised probe placed it first [lucas-overfitting-inverts-the-ranker-ordering]; on the suite the effect has the same direction at a magnitude of four ten-thousandths [suite-probe-gap-degrades-the-ordering]. A ranking comparison should therefore be reported with the probe's generalisation gap. Filtering a reference to large homogeneous units favours the ranking (AUROC 0.774 on the kept LUCAS units, 0.677 on the deleted ones) [lucas-the-homogeneity-filter-flatters-the-tool].

5 Negative results

No signal derived from the encoder's internals, its pretraining objective, a posterior over the probe head, feature-space typicality, a second model of the same family, or flip-and-rotate consistency ranks errors better than confidence on expert labels [no-encoder-internal-signal-beats-confidence] (Table 2). Choosing fine-tuning tiles by suspicion is worse than random selection at every budget from 100 to 1,000 labels [audit-does-not-save-labels].

Family	Result
Head ensembles, posteriors	A bootstrap bag's apparent gain did not replicate [bag-not-replicated]; Laplace variance is the worst signal on both testbeds [laplace-variance-rejected].
Internal states	Logit-lens settling, representation drift and attention entropy beat confidence on 0, 3 and 3 of 27 scenes [internal-state-signals-rejected].
Feature typicality	kNN, Mahalanobis, PCA-residual and ViM scores do not beat confidence [feature-typicality-rejected].
Second view	Two-view disagreement captures 0.242 of errors at 10% against 0.454 for confidence on Sen1Floods11 [two-view-disagreement-rejected]; disagreement with heads on other sensors beats the control [sensor-disagreement-beats-the-control] and does not yield a better review set [sensor-disagreement-does-not-pay-in-a-review-set].
Perturbation	Flip-and-rotate variation [dihedral-consistency-rejected] and masking [masking-perturbation-rejected] (no committed artifact) do not beat confidence.
Pretraining objective	The re-targeted residual tracks texture [retargeted-residual-rejected]; OlmoEarth v1's target has effective rank 2 [target-effective-rank-2].
Published signals	SHRUG-FM's signals [Gonzalez-Calabuig et al., 2025] do not beat confidence [shrug-signals-rejected].
Consensus	Within one family, Dawid–Skene [Dawid and Skene, 1979] inflates every member and inverts the order [family-errs-together-dawid-skene] (no committed artifact).

Table 2: Signal families that did not improve on the model's confidence.

exp77 tested a mechanism of arXiv:2602.22394 in which scene semantics diffuse into tokens. In the confident half of the windows, errors lie closer to their tile's mean token than correct windows on 5 of 7 tasks [confident-errors-are-scene-typical], a gap computed from labels without a class-frequency control; removing that direction costs 0.03 to 0.98 points of accuracy on all seven [removing-the-scene-direction-does-not-help], and scene typicality ranks errors worse than the margin on all seven [scene-typicality-loses-to-the-margin]. exp83 applied Dawid–Skene to encoders from different families. On Sen1Floods11, where eight encoders share 82% to 87% of their errors, it estimates 0.975 to 0.983 for maps 0.883 to 0.914 accurate, and its rank correlation with true accuracy is 0.83, 0.71 and 0.62 on PASTIS Sentinel-2, MADOS and Sen1Floods11 [consensus-order-recovered-only-where-true-gaps-are-large]. Within one family it estimates 0.996 to 0.999 for maps 0.81 to 0.93 accurate [within-family-consensus-is-the-unanimous-limit]. An error rate therefore requires a reference. Table 3 lists the failed predictions of exp77 to exp99.

6 Limitations

On published products, seven products against six reference samples have been graded (exp92 to exp94), five of the samples probability samples, with the points standing in for each map and the products compared through crosswalks; whether a reference point trained a product was not checked, and several pairs are related by construction or training; the rest of the labelled-route evidence is linear probes on frozen embeddings, graded against benchmark splits that are

not probability samples, and one fine-tuned model, FT-AWF, which is also the only model run through a deployment pipeline: one re-run graded at Ai2's expert-placed points (exp98) and one transfer test graded at the plots of a random sample that had input (exp99), both through the v1-era olmoearth_run configuration with runner and rslearn versions the record chose, not Ai2's lock or the current rslearn prediction. The ranking has been graded on four products' own confidence: Dynamic World's, LCMAP's, CGLS-LC100's and ODSE-LULC's. The hand-labelled testbeds are mostly binary water tasks, on which the margin, one minus the top probability and entropy coincide, and the fine-tuned evidence is FT-AWF's 41 errors on 344 points with Forest Loss Driver report-only [fine-tuned-model-audit], FT-AWF's re-run at 259 of those points, and its 2017 map at 271 TimeSync plots, whose missing plots are not missing at random and whose two crosswalks differ mostly at a legend boundary AWF does not define. exp98's equivalence interval and exp99's LENIENT AUROC and least-confident-tenth interval lie at or across their bars, and exp99 does not separate year, place, legend and point placement. The suite's 24 tasks come from 14 sources, its splits are the model authors' and not a landscape sample, and its controls are near chance. The estimation studies rest on seven segmentation tasks from five datasets, whose 64×64-pixel chip is smaller than a reviewer's unit, so the tile design effect is a lower bound. The shipped tile design under-covers where a few tiles hold most errors (84.3% on Sen1Floods11, 68.5% on MADOS), and the default confidence design's interval and the per-class intervals are nominal (14 of 628 per-class cells below 0.93, the lowest 0.879).

Every interval and certified zone describes agreement with the reviewer's labels. No experiment measured how often reviewers err; the reviewer-error bounds rest on rates the user states, and `certify` has no allowance for them. No real cloud has been tested: exp88's condition is whole-tile radar on three crop families, and a model trained with modality dropout is untested. Apart from exp88's missing input, ranking was measured on clean splits; no corruption or geographic shift of the kind Lehmann et al. [2026] apply was tested, and they report that confidence-based abstention cannot defer around confidently wrong predictions under such shift. No label-free reading tried has identified the errors a model makes confidently. The ranking carries no guarantee, and a review set is not a sample.

The agent evidence is binary water cards at two model sizes, a trial that passed only under an amended instrument, and a five-question pilot with no artifact. exp79's per-unit files are committed only for OlmoEarth Base's seed 0, so some crosschecks, including exp90's, cannot be rerun from the repository alone. The Lean proofs are not built in continuous integration, and the MCP server's tests run only in the release checks. Probe training was not bit-reproducible across GPU types, and the work has a single author.

7 Software

`olmoearth_inferenceX` is on PyPI as `olmoearth-inferencex` (Apache License 2.0); version 1.7.0 was released on 7 October 2026. Version 1.0.0 (17 September 2026) carried ranking, explanation, comparison and label-fitted fusion; 1.2.0 (22 September) added `sample` and `estimate`; 1.3.0 (24 September) added `certify`, per-class accuracy, dates in `compare` and the exact random-sample interval, and fixed the 33 findings of a second line-by-line review, each with a test that failed first. 1.4.0 (2 October) added input conditions, the MCP server, fixed-sequence wording for the prefix rule and the exact tie decision at a level; 1.4.1 rewrote the server's texts after the agent pilot; 1.5.0 added `?` windows, the reviewer-error bounds and the Lean proofs; 1.6.0 added the comparison of two maps from labels where they differ; 1.7.0 added `decide`, published products as input (a class map with its confidence band) and the fixes of a bug hunt in which seven finders, one per module group, reported 64 bugs, each shown by a script, and a second agent reproduced 60; each fix has a test, and two reviews of the fixes found that no recorded number changed. Since 1.4.1 each release has been preceded by a review in several lenses (from 1.5.0: statistics,

code, and documents with tests), whose findings were fixed with tests. For 1.7.0 the suite ran against the built wheel without the source tree on Python 3.11, 3.12 and 3.13 with the optional extras (1,392 passed each), on the declared minimum core dependencies (1,262 passed, the geo and MCP tests skipped), and, for the MCP server's and the new tests, on the lowest supported `mcp`. It includes enumeration checks of every interval and rule above, golden outputs of version 1.3.1 that later versions must reproduce byte for byte on maps without input conditions, apart from five listed changes, an added `scope` key and certify p-values within 10^{-12}, every command under every option, property tests on random maps, and every claim check. An OlmoEarth Agent skill ports the review set and agrees bit-exactly with the package's decisions on the suite's 17 single-label tasks [agent-skill-reproduces-the-package].

8 Data and code availability

Code, experiments, outputs (`exp/out/`), the claim ledger and this report are at `https://github.com/2imi9/olmoearth_inferenceX`: the package as released is at tag `v1.9.0`, and this report, with the documentation corrections made alongside it, is on the main branch after that tag. The Lean proofs and the experiments archived from the main branch on 10 October 2026, among them the agent trials exp86 and exp87 and their plans, are at tag `v1.9.0` (`https://github.com/2imi9/olmoearth_inferenceX/tree/v1.9.0`). Documentation is at `https://olmoearth-inferencex.readthedocs.io`; the per-experiment record is `docs/results/comparisons.md`. The data used are published by their providers: Sen1Floods11, GEOID-Flood, WorldFloods v2, DFC2020, Dynamic World, LUCAS Copernicus 2022, EuroCrops, LCMAP's Reference Data Product v1.2 and Collection 1.3 (USGS), Esri's 10 m Annual Land Use Land Cover v2, the OlmoEarth paper embeddings on Hugging Face (`allenai/olmoearth-paper-embeddings`) and Ai2's published fine-tuned checkpoints, with upstream revisions pinned in `exp/out/upstream_revisions.json`. Their licences differ (`docs/testbeds.md`): WorldFloods v2 is non-commercial, Sen1Floods11's labels carry no stated licence, and DFC2020 is released by the IEEE GRSS to approved contest participants, while the record read an unofficial mirror of it.

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Test	Predicted	Observed
exp77 P2; P3	gain ≥ 0.5 points on 4 of 7 tasks; gap ranks with gain	loss on 7 of 7; $\rho = -0.49$ [scene-contamination-diagnosis-has-nothing-to-locate]
exp78 P3; P4	ratio > 0.94 at over 30% error; saving $\leq 1.8\times$	0.9361; $2.51\times$
exp79 gate	seed-0 accuracies within 10^{-4}	missed by five encoders, on segmentation only [exp79-gate-fails-on-five-encoders-on-segmentation]
exp80 P2; P3	arithmetic agrees on every task; plug-in fails on 18 of 24	not on two near-census tasks [trust-zone-arithmetic-agrees-outside-census]; bar misstated
exp81 P1; P2	every cell ≥ 0.93; design wider on half the tasks	14 of 628 below; 2 of 21
exp81 P3	share test on every class	12 of 116 fail; holds from 30 expected labels, a rule set afterwards [adjusted-share-moves-with-the-population-gap]
exp82 P1; G1, G3	enrichment ≥ 2 up to ten classes; grain bars	1.24; missed by 0.004 and 5×10^{-5}
exp83 P1	hidden share within 0.10 of shared share	below it on 13 of 20 raters [consensus-hides-less-than-the-shared-share-and-debits-the-best]
exp83 P2; P3	rank correlation ≥ 0.8 on both diverse tasks; disagreement reads error	0.71 on MADOS; -0.03 on PASTIS [disagreement-reads-error-with-a-blind-spot]
exp84 P1	ρ(accuracy, AUROC) ≥ 0.4	positive everywhere, below 0.4 on 14 of 150 [exp84-confound-has-its-sign-not-its-size]
exp84 P3	sensor-group range of the lead ≤ 0.06	exceeded on 48 of 150; every group positive [exp84-lead-positive-in-every-sensor-group-but-uneven]
exp85 P3; P4	ratio ≤ 0.95; plain coefficient wider on two tasks	0.978; never wider [model-assisted-coefficient-is-one]; P5 withdrawn
exp86	answers pass under the preregistered instrument	pass only after fourteen amendments made after results were seen
exp88 P2 (China 6)	confident errors rise on CropHarvest China 6 without the optical input	confident share falls, 13.3% to 6.5% [missing-modality-confident-errors-follow-the-shift-not-the-sensor]
exp91 P1	a vision-language reviewer agrees with the hand labels on $\geq 90\%$ of judged windows	85.6% [exp91-vlm-reviewer-shares-the-maps-errors]
exp95 P2	the standard level cut's coverage falls by more than 0.05 when labels are added on at least 17 of 34 maps	14 of 34 [exp95-standard-cut-falls-when-labels-are-added]
exp98 P1	re-run within 5 points of the replica at Ai2's points	interval -5.1 to $+0.8$, 0.15 past the margin [exp98-equivalence-within-5-points-not-shown]
exp99 P1	design-weighted AUROC $\geq$ 0.70 on the TimeSync plots	0.638 STRICT, 0.678 LENIENT [exp99-confidence-misses-the-ranking-bar-under-both-rules]

Table 3: Preregistered predictions of exp77 to exp99 recorded as failed or failed as written. exp89 was report-only by an amendment made before any result, exp90 was not preregistered, and the predictions of exp92, exp93, exp94, exp96 (at its preregistered seed) and exp97 held.